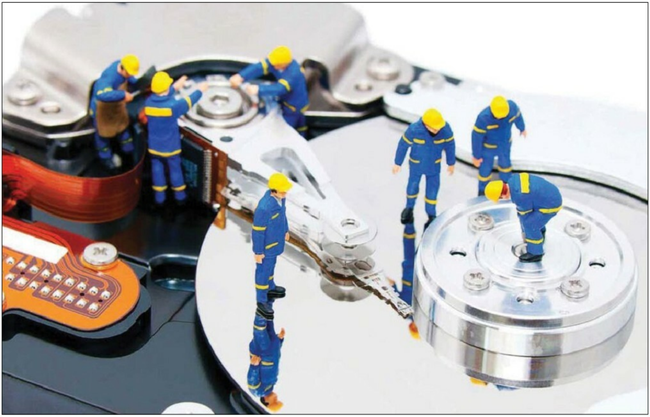
An Introduction to Distributed Replicated Block Device

Distributed Replicated Block Device (DRBD) mirrors block devices among multiple hosts. The replication is visible to other applications on the host systems. Block devices such as hard disks, partitions, logical volumes, etc, can be mirrored to form high availability (HA) clusters.



In this article, let's explore some of the technology that works behind the cloud. To set up enterprise level cloud computing infrastructure, we not only need server grade hardware and cloud software but also something more. To ensure continuous availability and integrity of user data and application resources, we need a storage solution which can protect service continuity from individual server failures, network failures or even natural disasters. DRBD or Distributed Replicated Block Device is an open source solution that provides a highly available, efficient, distributed storage system for the cluster environment. It is developed and maintained by LINBIT, Austria and is available under GPLv2.

Architecture

Let's understand the DRBD architecture with the help of an illustration (Figure 1) taken directly from the DRBD documentation. The two boxes represent two DRBD nodes forming a high availability (HA) cluster. A typical Linux distribution contains a stack of modules which more or less include a file system, a buffer cache, a disk scheduler and a disk driver to handle storage devices. The DRBD module sits in the middle of this stack, i.e., between the buffer cache and the disk scheduler.

The black arrow represents a typical flow of data in response to the normal application disk usage. The orange arrow represents data flow initiated by DRBD itself for replication purposes, from one node to another via the network. For the purposes of our discussion, let's assume the left node to be the primary and the right node to be the secondary DRBD node. Whenever a new disk block